External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask what status evidence proves and which separate authority any next action needs.

Use unchanged committed records, conditional same-key resend candidates and separately approved compensation plans without claiming real recovery execution.

Keep one habit: Model a lost acknowledgement → inspect status distinctions → compare reliable/unreliable absence → assess fresh unknown-outcome reasoning → preserve reconciliation history.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of external side effects and recovery

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of timeouts retries and cancellation

Reference example, not the only acceptable wording: Teacher explanation: Timeouts bound waiting; retries consume a finite budget. Cancellation should stop future work while accounting for operations already committed.

Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of planning and action selection

Reference example, not the only acceptable wording: Teacher explanation: Planning proposes steps; execution must still validate permissions and prerequisites for each step. Replanning cannot invent authority.

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of failure budgets and service objectives

Reference example, not the only acceptable wording: Teacher explanation: Service objectives define measured reliability over a stated interval.

A failure budget is meaningful only after success and eligible requests are defined. For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are 2%*200=4.

Count every eligible failure once; report excluded requests separately.

Pictured end goal

Explain external side effects and recovery using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Timeouts retries and cancellation

Planning and action selection

Failure budgets and service objectives

Use the three-session teaching cycle above, with extra time as needed. External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of external side effects and recovery

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach status reconciliation before retry or compensation

> Separate trusted-fixture assumptions, pure decisions and real external actions.

Definitions and complete status/authority contract

External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

Recovery fixture or requirement	Complete local meaning
Original operation ID	BOOK-1, kept stable during recovery
Committed fixture	Authoritative matching status; result booking B1 already exists
Prepared fixture	Work was prepared; not proof it failed or is safe to repeat
Unknown/unavailable fixture	Outcome unresolved; review/reconcile rather than assume failure
Reliable absent fixture	Different evidence from silence; still needs separate resend authority and provider duplicate controls
Status access	Trusted permission to inspect the operation is assumed or denied by a supplied fixture flag
Compensation	A separate proposed cancel-booking operation COMP-1, with its own exact approval fixture
Code scope	Pure local decision helper; no status provider, booking, printer or token-transfer call

Main recovery map and checked reasoning

Reconcile the original operation before any new side effect

RELIABLE MATCHING EVIDENCE	RECOVERY DECISION	WHAT MUST NOT HAPPEN
BOOK-1 committed, existing booking B1	Return the existing result to an authorized caller	Do not create a second booking
BOOK-1 prepared	Pending review / inspect authoritative progress	Do not call preparation a confirmed failure
BOOK-1 unknown or status unavailable	Keep outcome unresolved and reconcile	Do not repeat blindly with a fresh key
Status is for another ID or is unreliable	Review the evidence mismatch	Do not report it as BOOK-1's confirmed result
Proposed compensation	Check separate exact authority and preserve original history	Do not silently cancel/delete/refund as part of reading status

The helper starts from supplied status fixtures and creates no effects. A recovery decision and an executed recovery action are separate stages.

Require stable operation identity and reliable authorized status before declaring an outcome. The main booking already exists; returning its result creates no new booking. Prepared is not failed, unknown is not absent, and missing acknowledgement is not proof of rollback.

The printer’s reliably absent case may produce a separately approved same-key resend candidate only with required provider controls; it is not an executed resend. Compensation has separate authority and a separate operation identity. The mock trusts supplied access/evidence flags and an exact approval fixture; it does not implement their real verification or a durable recovery protocol.

The independent token-transfer scenario remains entirely fictional and should not be repeated blindly.

External side effects and recovery — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A booking is committed in storage, but response delivery times out. Caller asks to repeat.	The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action.	Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A booking is committed in storage, but response delivery times out. Caller asks to repeat.

Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Reliable evidence, read-access and separate-effect probes

Recovery probe	Expected decision or limit
Committed matching status with a valid result	Return existing copied result, no new effect
Prepared	Pending review; do not infer confirmed non-execution
Unknown or unavailable	Reconcile, no blind repeat
Reliable absence but no resend approval	Review; evidence does not authorize the action
Resend approval but unreliable status	Review; permission does not supply missing evidence
Reliable absence + trusted resend approval + provider deduplication	Return a same-key resend candidate, not evidence it was executed
No status-read permission	Deny disclosure under the supplied policy fixture
Compensation requested without its own authority	Reject that separate action; preserve original committed record
Pure local recovery checks pass	No real provider consistency, durable recovery or exactly-once guarantee established

Separate printer job and conditional-resend scope

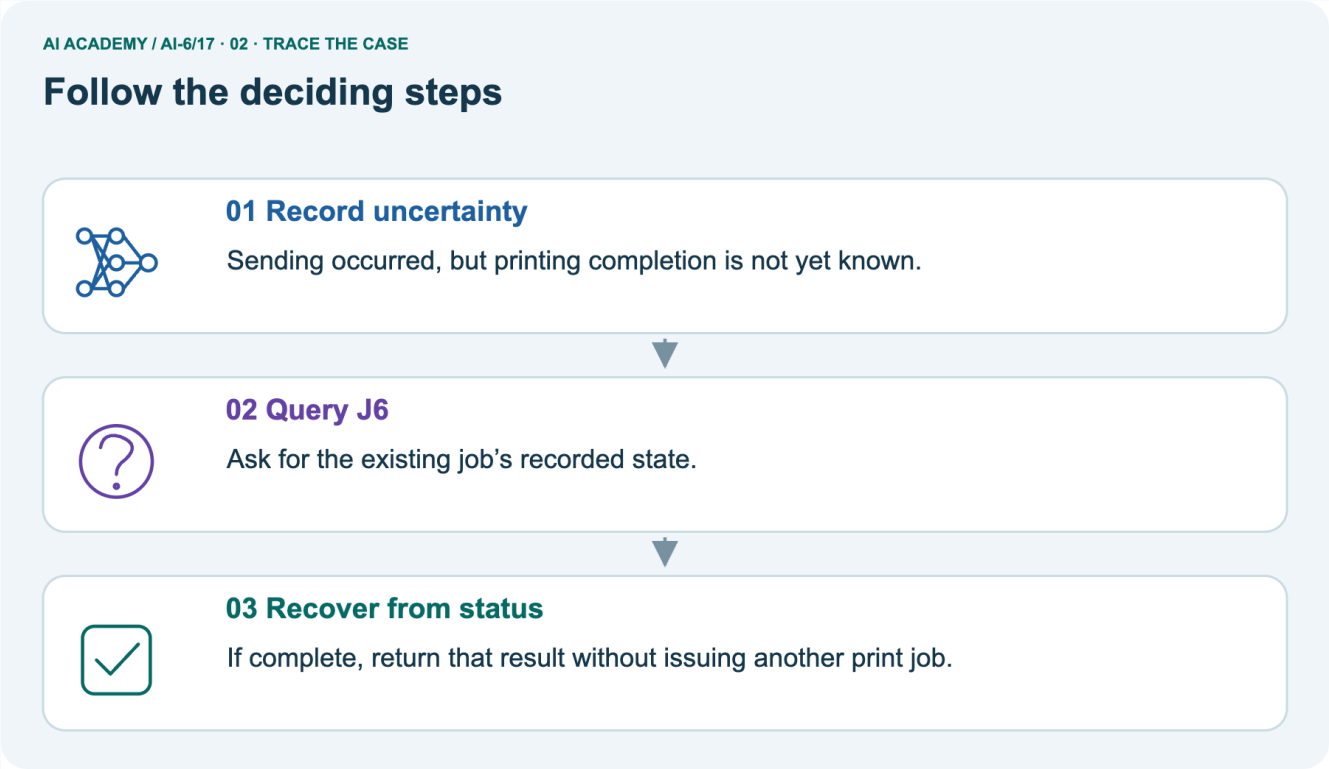
Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

The main booking uses BOOK-1/B1; the picture is a separate printer job J6 for one fictional badge. A missing reply leaves printing outcome unknown. An authorized status lookup that reliably reports J6 complete supports returning the existing result without another print.

Reliable absence is different from silence, but only a separately approved same-key resend with the required provider duplicate controls may be considered. If those controls or reliable status are missing, keep review/uncertainty. No print command is executed in this lesson.

Printer trace and reliably absent-status key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record uncertainty	Sending occurred, but printing completion is not yet known.
2. Query J6	Ask for the existing job’s recorded state.
3. Recover from status	If complete, return that result without issuing another print job.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Change: Status reliably reports J6 absent.

Prediction: A policy-approved resend using the same deduplication key may be considered.

Reason: Confirmed absence supplies different **evidence** from silence.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Acknowledgement	A response confirming operation status
Reconciliation	Checking authoritative state after uncertainty
Compensation	A separate action intended to offset a prior effect

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

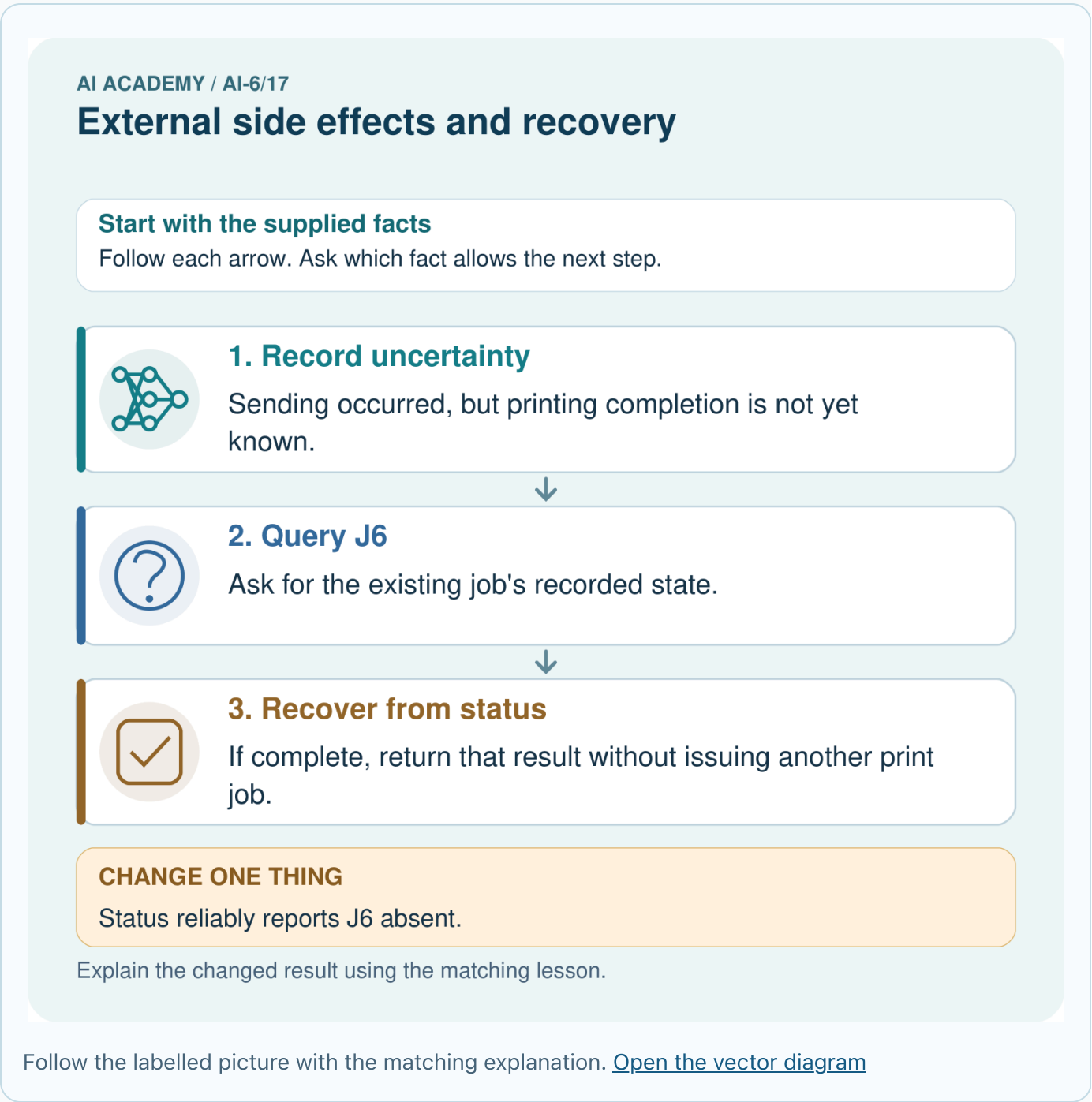
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: External side effects and recovery

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Sending occurred, but printing completion is not yet known. Ask for the existing job's recorded state. If complete, return that result without issuing another print job.

Changed case: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Changed-case reasoning: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main committed booking returns existing result, no second booking. Prepared/unknown remain unresolved; reliable absent printer J6 only permits consideration of a separately authorized same-key resend under provider controls. Compensation needs separate authority and preserves original commit.

Independent unknown token-transfer status after network loss must not repeat immediately: keep stable operation identity, inspect reliable authorized status, reconcile or review uncertainty, and apply duplicate/approval controls before any separately permitted new action. No real money, transfer or external action is involved.

Diagnose the current misconceptions

The worked result establishes every future case: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

Teacher explanation: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

W2 Trace the worked case

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

W3 Audit the limit

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

Teacher explanation: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only. Require intermediate reasoning and the stated scope.

Fresh retry: A send ledger says committed while client reports timeout. Expected reasoning: Treat the committed record as effect **evidence**; return its result and avoid duplication.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Untrusted **input** boundaries. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/18, ai-6/20, ai-6/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of external side effects and recovery

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Check every learner before discussion: Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Status reliably reports J6 absent. Require a **prediction** and a reason before comparing. Expected result: A policy-approved resend using the same deduplication key may be considered.

Reason: Confirmed absence supplies different **evidence** from silence.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

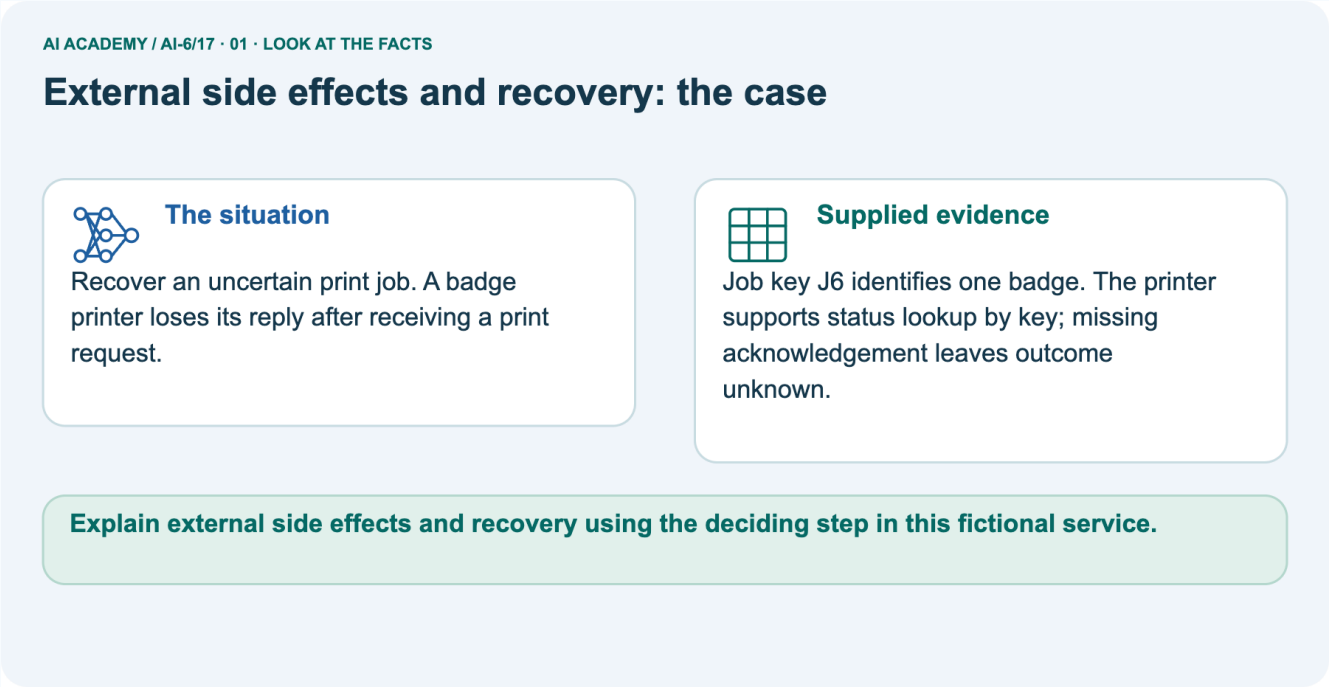
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. External side effects and recovery: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

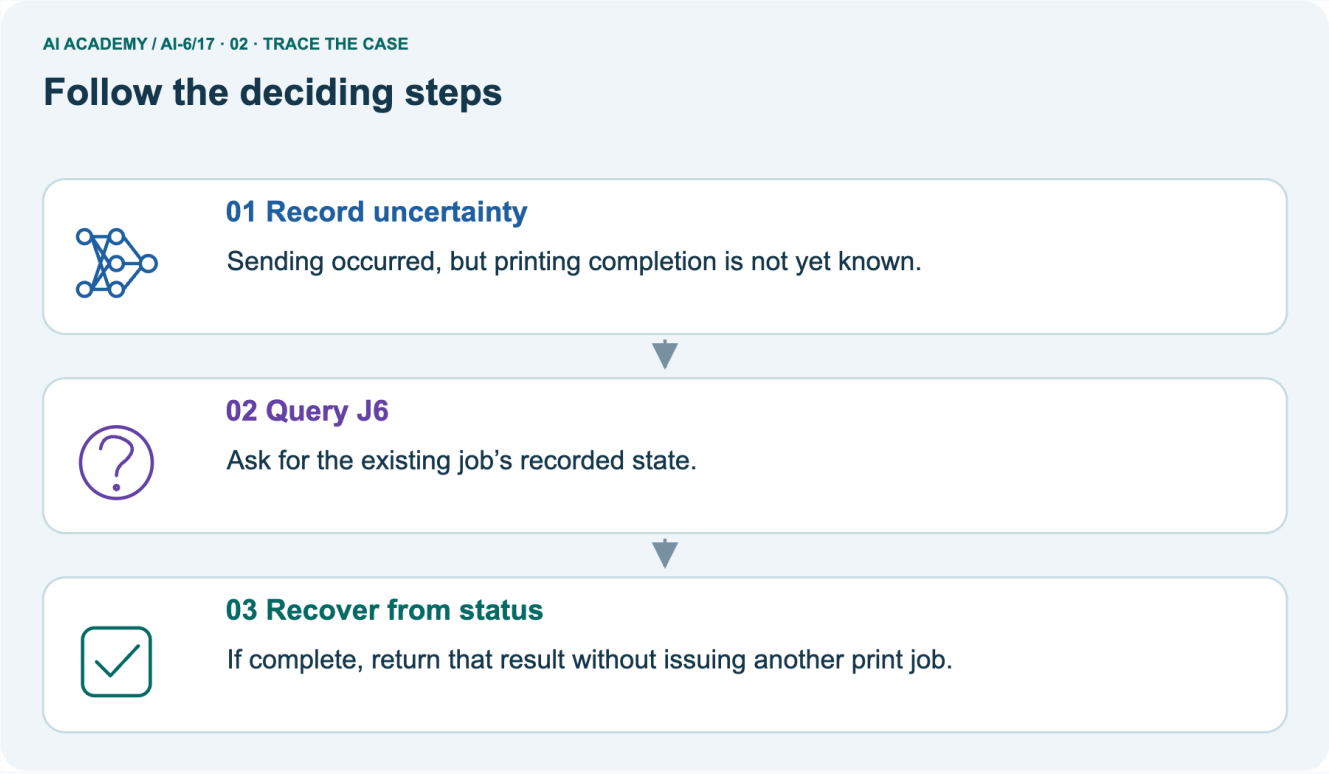
Recover an uncertain print job. A badge printer loses its reply after receiving a print request. Supplied facts: Job key J6 identifies one badge.

The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



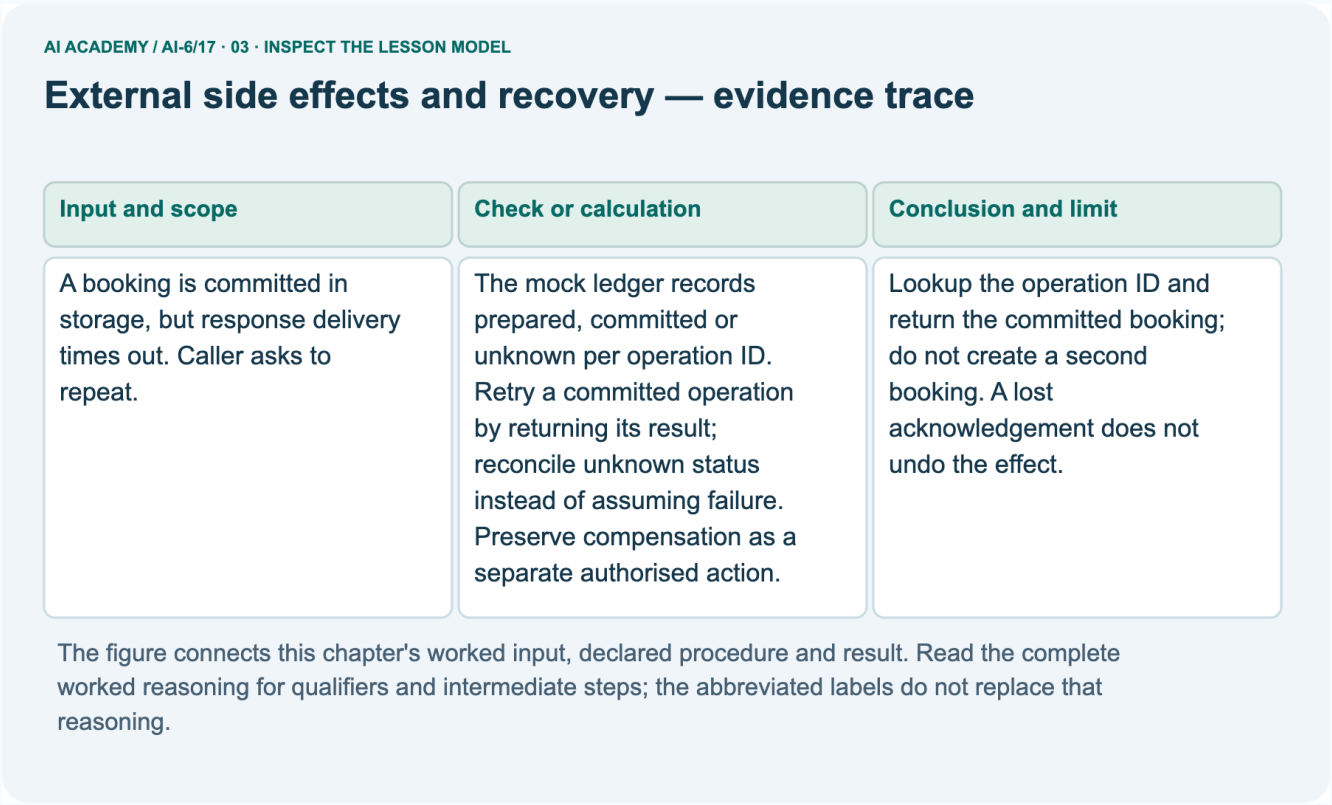
Follow the labelled picture with the matching explanation. Open the vector diagram

Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. External side effects and recovery — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

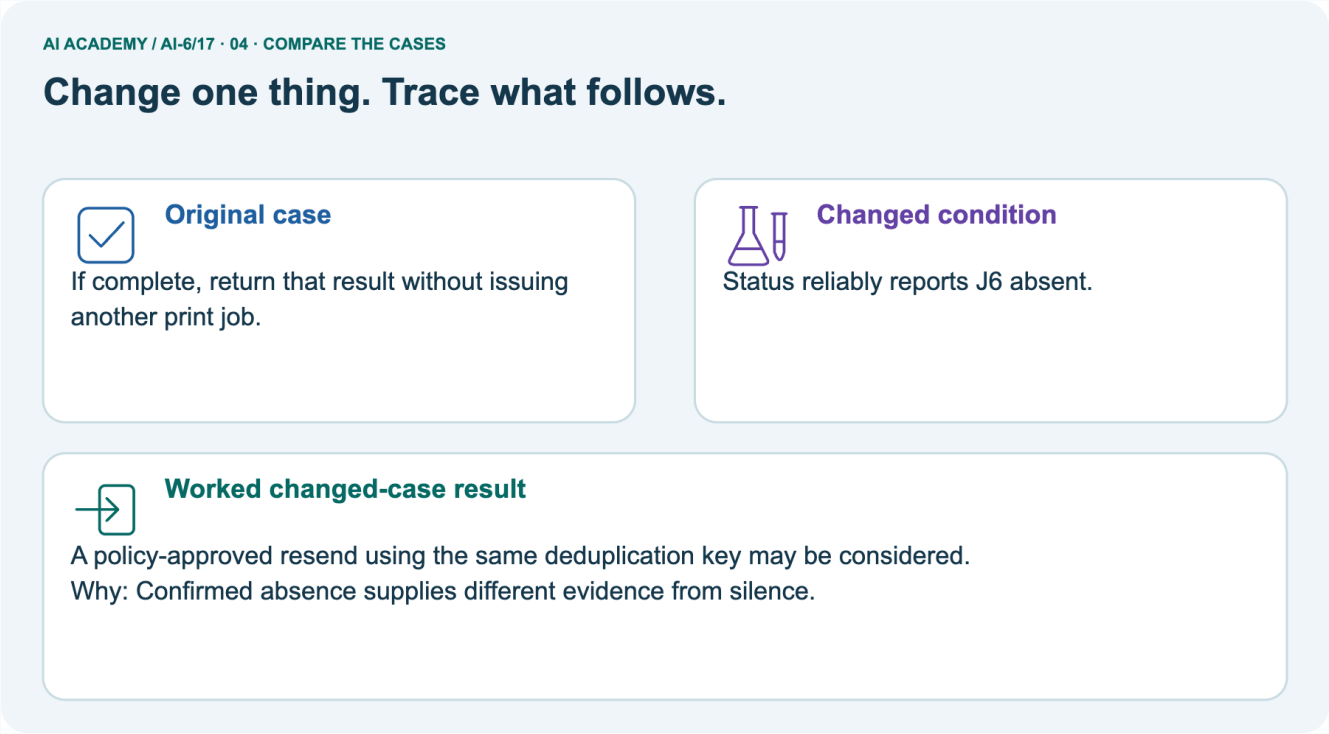
Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

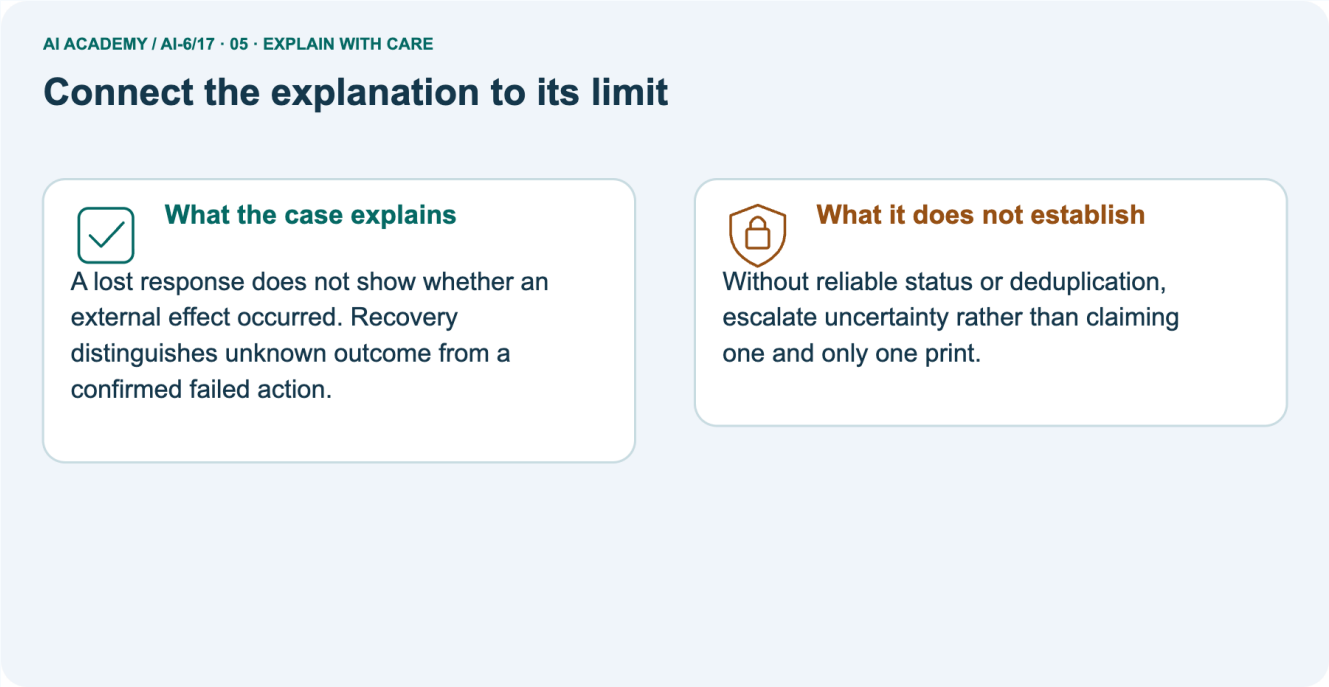
Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Confirmed absence supplies different **evidence** from silence.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

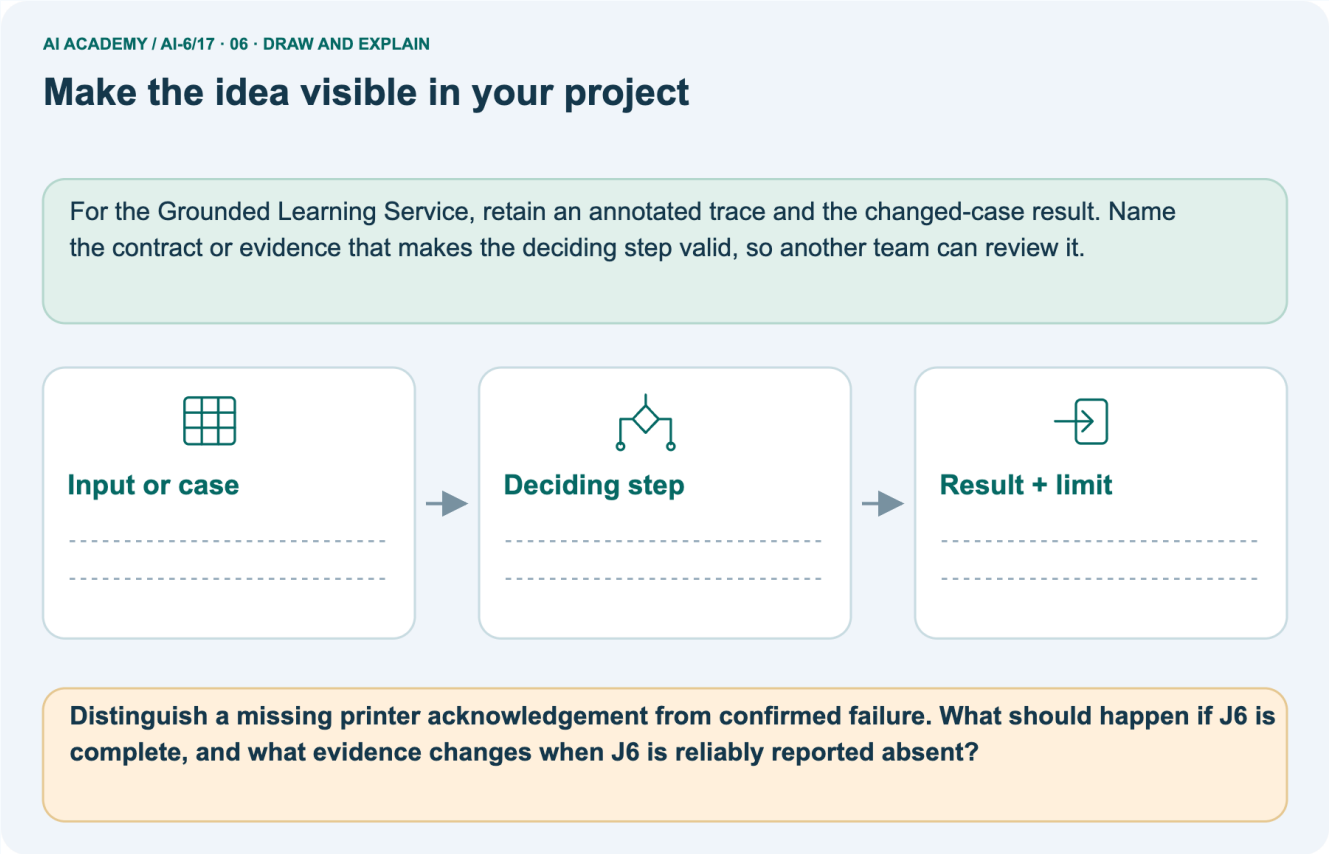
Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Sending occurred, but printing completion is not yet known. Ask for the existing job's recorded state. If complete, return that result without issuing another print job.

Changed case: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. External side effects and recovery: the case

02. Follow the deciding steps

03. External side effects and recovery — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.